SINT Protocol: A Runtime Authorization Framework for Physical AI Agents

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Open source: github.com/sint-ai/sint-protocol

Abstract

Physical AI agents — LLM-driven systems controlling robots, drones, and actuators — introduce a safety failure mode that neither classical robot safety (IEC 62443, ISO 10218) nor LLM alignment research fully addresses: **an authenticated, aligned agent issuing a physically irreversible command without appropriate human oversight, at the wrong moment, in a degraded environment.**

We present **SINT Protocol**, an open-source runtime authorization framework that enforces six safety invariants at every agent action boundary.

Safety Invariants

1. Capability confinement. Every action is authorized by a signed, expiring Ed25519 capability token scoped to a specific resource, action set, and physical constraint envelope (max velocity, max force, geofence polygon). Tokens cannot be self-issued or silently expanded; delegation is *attenuation-only* with cryptographic enforcement.

2. Tier-based human oversight. Actions are classified into four approval tiers: `T0_observe` (auto-allow), `T1_prepare` (audit), `T2_act` (human review), `T3_commit` (explicit sign-off with M-of-N quorum). Classification uses 50+ pattern-matching rules covering ROS 2 topics, MAVLink commands, and MCP tool calls. Shell and code-execution tool names (`bash` , `exec` , `eval`) are explicitly classified at `T3_commit` (ASI05).

3. Behavioral drift detection (CSML). The Composite Safety-Model Latency score tracks an agent's out-of-policy action rate, timing anomalies, and safety event frequency across a sliding evidence window. When CSML exceeds threshold θ, the agent's tier is automatically escalated — making behavioral drift a *live enforcement trigger* rather than a post-incident finding. Frontier LLMs exhibit up to 4.8× differences in out-of-policy action rates on identical tasks [Cardenas et al., 2026]; SINT makes this variance enforcement-visible.

4. Environment-adaptive constraint tightening. Token constraints are necessarily static at issuance. The `DynamicEnvelopePlugin` maps real-time sensor state to tighter effective limits: `v_eff = min(token.maxVelocityMps, d_obs × r_f)` . This closes the gap between "configurable safety envelope" and "environment-adaptive safety envelope" identified in ROSClaw [Cardenas et al., 2026].

5. Active safety plugins. The `GoalHijackPlugin` (ASI01) provides 5-layer heuristic detection: prompt injection, role override, semantic escalation, exfiltration probes, and cross-agent injection. The `MemoryIntegrityChecker` (ASI06) detects replay attacks, privilege claims, history overflow, *cross-session continuity injection* (hallucinated prior-auth claims), *credential read funnels* (≥3 secret reads followed by a write), and *action velocity loops* (last 5 actions targeting an identical resource — hallucination loop indicator). The `DefaultSupplyChainVerifier` (ASI04) validates model fingerprint hashes at runtime to detect tampered tools or swapped model versions.

6. Stop button (EU AI Act Art. 14(4)(e)). The `CircuitBreakerPlugin` allows operators to manually `trip()` the circuit, instantly blocking all actions from a rogue agent. N consecutive denials auto-open the circuit; CSML anomalous-persona detection auto-trips the circuit; HALF_OPEN probe recovery ensures self-healing after incident resolution.

Swarm Safety

For multi-agent deployments, `SwarmCoordinator` enforces collective constraints before individual gateway calls: maximum concurrent actors in T2+ tier, total kinetic energy ceiling ($\Sigma \frac{1}{2}mv^2$), minimum inter-agent distance, and maximum escalated fraction. These collective constraints cannot be captured by per-agent token scoping alone.

Operator Interface Layer

SINT's governance-first operator layer comprises: (i) `@sint/memory` — a ledger-backed operator memory system where every recall emits a typed `SintLedgerEvent`; (ii) a voice-first SINT Command HUD routing every voice command through `PolicyGateway.intercept()` before execution; (iii) a `ProactiveEscalationEngine` that monitors CSML drift across multi-turn windows and emits operator notifications at $\theta = 0.3$; (iv) hierarchical token delegation (`sint__delegate_to_agent` , max depth 3) with atomic cascade revocation via DFS subtree traversal; and (v) a REST API surface (`/v1/memory` , `/v1/delegations` , `/v1/csml`) for governance UIs.

Implementation & Compliance

SINT is implemented in TypeScript across **41 packages**, with **1,772 tests** covering core, gateway, bridges, and the avatar layer. Bridge adapters exist for ROS 2, MAVLink v2, MCP tool calls, Google A2A, MQTT/CoAP IoT, OPC-UA, Open-RMF, and Sparkplug B. The Evidence Ledger implements a SHA-256 hash chain with optional TEE attestation (Intel SGX, ARM TrustZone, AMD SEV) for post-incident forensics.

SINT covers all 10 OWASP Agentic Top 10 categories (ASI01–ASI10). Coverage is formally declared in a machine-readable conformance mapping (`docs/conformance/owasp-asi-mapping.md`) and regression-tested across 29 attack/safe-case fixture pairs that validate each ASI category against the gateway choke point.

Position

Safe physical AI requires that authorization be treated as a first-class *runtime* concern — not a deployment configuration or a prompt engineering problem. The safety properties that matter (who approved this action, was the envelope respected, was the agent's drift within bounds) must be cryptographically auditable and enforced *before* hardware execution, not after. SINT Protocol is a concrete, open-source instantiation of this position.

References

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